

Project Title:

Login System Testing using Manual and Automation Testing Techniques

1. Introduction

Software testing is the process of checking whether a software application works correctly or not. It helps identify errors, defects, and missing requirements before the software is delivered to users.

Testing improves software quality, reliability, and performance. In this project, both manual testing and automated testing techniques are applied to a web-based login authentication system.

The project also demonstrates the use of Selenium IDE for automation testing.

2. Objective

The objective of this project is:

- To understand software testing techniques
- To perform manual testing using black box testing
- To apply Equivalence Class Partitioning and Boundary Value Analysis
- To perform automated testing using Selenium IDE
- To identify defects in the system
- To compare manual and automation testing approaches

3. Selected System

Login Authentication System

The selected system is a web-based login authentication system used to verify user credentials. The system allows users to enter a username and password and grants access only when valid credentials are provided.

4. Modules of System

Module	Description
Login Module	Allows users to login
Username Validation	Checks username validity
Password Validation	Checks password validity
Authentication Module	Verifies credentials
Error Handling Module	Displays validation messages

5. Test Plan

5.1 Objective

To verify that all modules of the login authentication system work correctly according to requirements.

5.2 Scope

The testing includes:

- Valid login functionality
- Invalid password validation
- Empty username validation
- Empty password validation
- Invalid username validation

5.3 Testing Techniques

The following testing techniques are used:

- Black Box Testing
- Equivalence Class Partitioning
- Boundary Value Analysis
- Automation Testing using Selenium IDE

5.4 Entry Criteria

Before testing:

- System should be installed
- Browser should be working
- Internet connection should be available
- Test data should be prepared

5.5 Exit Criteria

Testing will stop when:

- All test cases are executed
- Major defects are identified
- Test results are documented

6. Manual Testing

Manual testing is performed without using automation tools.

In this project, black box testing is used where only input and output behavior is checked.

6.1 Manual Test Cases

TC ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
TC01	Login	Valid Login	Correct username and password	User should login successfully	Login successful	Pass
TC02	Login	Invalid Password	Correct username and wrong password	Error message should appear	Error message displayed	Pass
TC03	Login	Empty Username	Empty username	Validation message should appear	Validation displayed	Pass
TC04	Login	Empty Password	Empty password	Validation message should appear	Validation displayed	Pass
TC05	Login	Invalid Username	Wrong username	Error message should appear	Error message displayed	Pass
TC06	Login	Valid Username Format	Correct username format	Username accepted	Username accepted	Pass
TC07	Login	Password Field Validation	Empty password field	Error should display	Error displayed	Pass
TC08	Login	Authentication Check	Correct login credentials	Dashboard should open	Dashboard opened	Pass

7. Equivalence Class Partitioning

Equivalence Class Partitioning divides input data into valid and invalid groups.

Example: Username Field

Input	Class
student	Valid
admin123	Valid
blank	Invalid
@@@@	Invalid

8. Boundary Value Analysis

Boundary Value Analysis checks values at the edges of input limits.

Example: Username Length

Allowed length = 5 to 15 characters

Input Length	Result
4	Invalid
5	Valid
6	Valid
14	Valid
15	Valid
16	Invalid

9. Automation Testing using Selenium IDE

9.1 Introduction to Selenium IDE

Selenium IDE is an automation testing tool used for recording and replaying web application tests.

It is easy to use because coding is not required.

9.2 Installation Steps

1. Open Mozilla Firefox
2. Search Selenium IDE extension
3. Install Selenium IDE
4. Open Selenium IDE
5. Create new project
6. Enter website URL
7. Start recording tests

10. Automated Test Cases

Automated Test Case 1 – Valid Login

Steps

1. Open browser
2. Enter correct username
3. Enter correct password
4. Click login button

Expected Result

User should login successfully.

Actual Result

Login successful.

Status

Pass

Automated Test Case 2 – Invalid Password

Steps

1. Open login page
2. Enter correct username
3. Enter wrong password
4. Click login button

Expected Result

Error message should display.

Actual Result

Error message displayed.

Status

Pass

Automated Test Case 3 – Empty Username

Steps

1. Open login page
2. Leave username field empty
3. Enter password
4. Click login button

Expected Result

Validation message should display.

Actual Result

Validation message displayed.

Status

Pass

Automated Test Case 4 – Empty Password

Steps

1. Open login page
2. Enter username
3. Leave password field empty
4. Click login button

Expected Result

Validation message should display.

Actual Result

Validation message displayed.

Status

Pass

Automated Test Case 5 – Invalid Username

Steps

1. Open login page
2. Enter invalid username
3. Enter correct password
4. Click login button

Expected Result

Error message should display.

Actual Result

Error message displayed.

Status

Pass

11. Defect Reporting

During testing, no critical functional defects were observed. However, the following minor issues and potential improvement areas were identified.

Defect ID	Description	Severity	Affected Module
D01	Generic error message is displayed for different login failures	Low	Authentication Module
D02	User interface can be improved for better usability and clarity	Low	Login Module
D03	Login response time can be optimized further under heavy load	Low	Authentication Module

12. Comparison of Manual and Automated Testing

Manual Testing	Automated Testing
Performed by humans	Performed using tools
Slow process	Fast process
Good for small systems	Good for large systems
Requires more time	Saves time
Repetitive tasks are difficult	Repetitive tasks are easy

Screenshots Section

1. Valid Login Screenshot – PASS

The screenshot displays the Selenium IDE interface for a project named 'P1*'. The test suite 'T1*' is selected, and the 'Run current test' button is visible. The test steps are as follows:

Step	Command	Target	Value
6	✓ execute script	document.getElementById('password').value='Password123';	
7	✓ click	id=submit	
8	✓ click	css=.post-title	
9	✓ assert title		Logged in Successfully Practice Test Automation

The status bar indicates 'Runs: 1 Failures: 0'. The log shows the following steps:

Log	Reference
4. executeScript on document.getElementById('username').value='student'; OK	22:42:54
5. click on id=password OK	22:42:54
6. executeScript on document.getElementById('password').value='Password123'; OK	22:42:54
7. click on id=submit OK	22:42:54
8. click on css=.post-title OK	22:42:54
9. assertTitle on Logged In Successfully Practice Test Automation OK	22:42:55

2. Invalid Password Screenshot – PASS

The screenshot displays the Selenium IDE interface for a project named 'P1*'. The test suite 'Invalid Login Test*' is selected, and the 'Run current test' button is visible. The test steps are as follows:

Step	Command	Target	Value
6	✓ execute script	document.getElementById('password').value='wrong123';	
7	✓ click	id=submit	
8	✓ assert text	id=error	Your password is invalid!

The status bar indicates 'Runs: 1 Failures: 0'. The log shows the following steps:

Log	Reference
3. click on id=username OK	23:00:22
4. executeScript on document.getElementById('username').value='student'; OK	23:00:25
5. click on id=password OK	23:00:25
6. executeScript on document.getElementById('password').value='wrong123'; OK	23:00:25
7. click on id=submit OK	23:00:25
8. assertText on id=error with value Your password is invalid! OK	23:00:25

3. Invalid Username Screenshot – PASS

Extension: (Selenium IDE) - Selenium IDE - P1* — Mozilla Firefox

Project: P1*

Tests +

Search tests...

✓ Empty Username Test*

✓ Invalid Login Test*

✓ T1*

https://practicetestautomation.com

	Command	Target	Value
4	✓ execute script	document.getElementById("username").value="";	
5	✓ click	id=password	
6	✓ execute script	document.getElementById("password").value="Password123";	
7	✓ click	id=submit	
8	✓ assert text	id=error	Your username is invalid!

Command

Target

Value

Description

Log Reference

2. Selenium IDE on 12/04/2023 OK 23:04:17

3. click on id=username OK 23:04:17

4. executeScript on document.getElementById("username").value=""; OK 23:04:19

5. click on id=password OK 23:04:19

6. executeScript on document.getElementById("password").value="Password123"; OK 23:04:20

7. click on id=submit OK 23:04:20

8. assertText on id=error with value Your username is invalid! OK 23:04:20

4. Empty Username Screenshot – PASS

Extension: (Selenium IDE) - Selenium IDE - P1* — Mozilla Firefox

Project: P1*

Tests +

Search tests...

✓ Empty Username Test*

✓ Invalid Login Test*

✓ T1*

✓ Invalid Password Test*

✓ Invalid Username Test*

https://practicetestautomation.com/

	Command	Target	Value
3	✓ click	id=username	
4	✓ execute script	document.getElementById("username").value="";	
5	✓ click	id=password	
6	✓ execute script	document.getElementById("password").value="Password123";	
7	✓ click	id=submit	

Command

Target

Value

Description

Log Reference

2. Selenium IDE on 12/04/2023 OK 00:09:29

3. click on id=username OK 00:09:31

4. executeScript on document.getElementById("username").value=""; OK 00:09:31

5. click on id=password OK 00:09:31

6. executeScript on document.getElementById("password").value="Password123"; OK 00:09:31

7. click on id=submit OK 00:09:31

8. assertText on id=error with value Your username is invalid! OK 00:09:31

5. Empty Password Screenshot – PASS

The screenshot displays the Selenium IDE interface within a Mozilla Firefox browser window. The project is named 'P1*'. The test suite 'Tests' is expanded, showing a list of test cases. The 'Empty Password Test*' is selected, and its details are shown in the main pane. The test consists of 8 steps:

Step	Command	Target	Value
4	execute script	document.getElementById('username').value='student';	
5	click	id=password	
6	execute script	document.getElementById('password').value='';	
7	click	id=submit	
8	assert text	id=error	Your password is invalid!

Below the test steps, there is a section for adding new steps with fields for Command, Target, Value, and Description. At the bottom, the 'Log' tab is active, showing a list of log entries with timestamps:

Log	Reference	Time
2. Setting browser to Firefox 4.0.1		00:13:22
3. click on id=username OK		00:13:23
4. executeScript on document.getElementById('username').value='student'; OK		00:13:24
5. click on id=password OK		00:13:24
6. executeScript on document.getElementById('password').value=''; OK		00:13:25
7. click on id=submit OK		00:13:25
8. assertText on id=error with value Your password is invalid! OK		00:13:25

13. Conclusion

In this project, both manual and automation testing techniques were applied to a web-based login system. Manual testing was performed using Black Box Testing, Equivalence Class Partitioning, and Boundary Value Analysis to verify different input conditions and improve test coverage.

Automation testing was carried out using Selenium IDE. Multiple test cases, including valid login, invalid password, invalid username, empty username, and empty password scenarios, were executed successfully. The automation process reduced testing effort and provided quick and reliable results.

Based on the testing performed, the login functionality behaved as expected and validation messages were displayed correctly for invalid inputs. This project demonstrated that manual testing is useful for understanding system behavior, while automation testing is more efficient for repetitive test execution.

14. References

1. Selenium IDE Documentation
2. Software Testing Lecture Notes
3. Web-Based Login System Demo
4. Software Testing Concepts and Techniques